

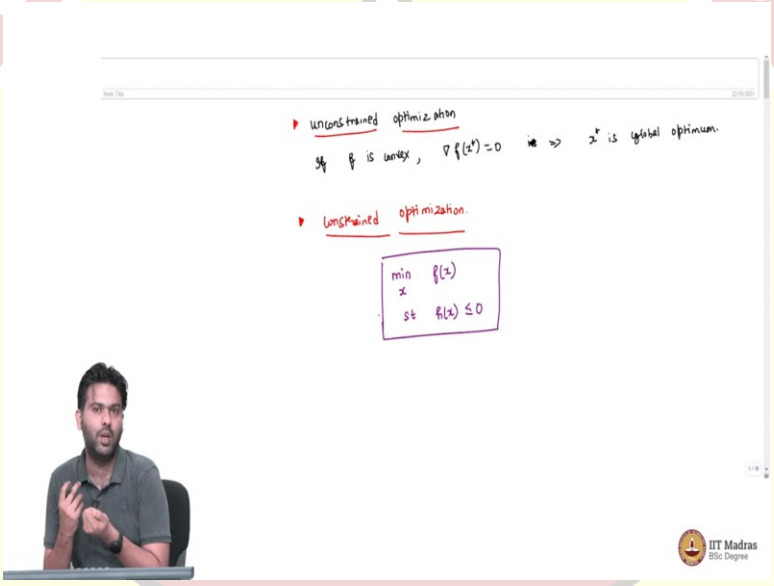
IIT Madras

ONLINE DEGREE

MLF
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Revisiting Constrained Optimization

Welcome back everyone. So today, we will look at starting, sorry about that; I will start again. Welcome back everyone. So today we will continue our discussion about optimization, and in particular what we have seen so far is the following.

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So, we have looked at unconstrained optimization so far. Specifically, we looked at a very interesting result, which said that if the function that you want to optimize is convex; if f is convex. And you are, and if there are no constraints; then, the gradient del f of x* equal to 0. So, the gradient at x* if it is 0, which means that this is what is called as a first order optimality condition, which means that the.

If you find the point x* where the gradient of the function is 0; then that condition is enough to say that x* is indeed a global optimum of the function s of the function f. 0 in place, x* is global optimum. And we know how to find that such an x*; so, we know how to algorithmically find such an x*. In particular, we can run our favourite gradient decent algorithm, and that would get us to an x*, which is optimum.

Of course, there maybe be other algorithms which will exploit the fact that f is convex and so on. But then, that is not the goal of this exercise; all we want to see is that; if the gradient is 0, then it is global optimum. Which means that you have an algorithm, which can find point where the gradient is 0; and so that will give you global optimum; so, but, note that all this worked for the unconstrained case. So, if f is convex, the unconstrained optimization problem seems easy.

Now, what we want to focus on today is the constrained optimization case. So, this is the constrained optimization problem that we want to look at, which we have already looked at in some detail. But then, we did not formalize the necessary or a sufficient condition for a point to be an optimal solution. So, we are going to revisit this constrained optimization problem today. And what is a constrained optimization problem?

We will start with a simple case; and then I will kind of hint at what would be the easy way to look at a generalized version of whatever. We will see for the simple case; so, we want to minimize over x, let us say x is Rn f of x. And now we have a constrained; let us call this constrained h of x is ≤ 0. So, this is a standard, the simple only one inequality constrained optimization problem.

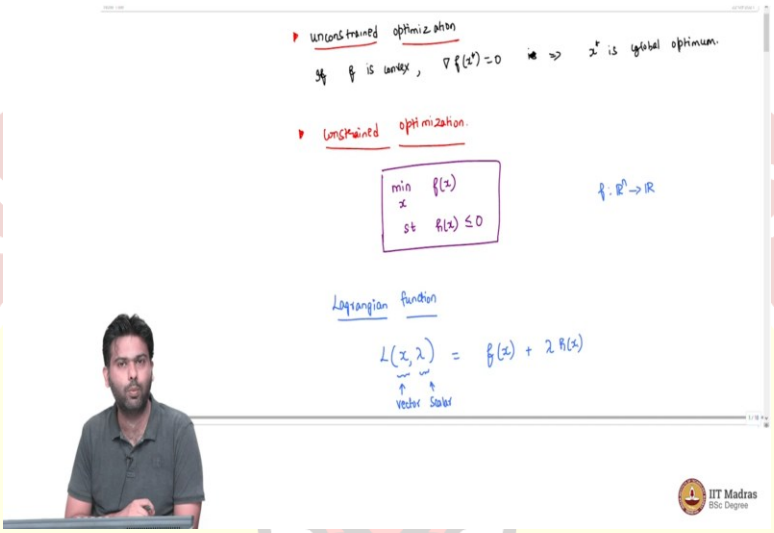
Now, what we are after is to find some kind of a sufficient condition for a given point to be; let us say local minima of this function. Or, if you assume more things about the functions case can we find the necessary condition for a point to be the minima of this function, and so on. So, these are the type of questions that we want to ask. So, so note that just as a small point that I want to mention here is that, so the theory of constrained optimization is quite rich and very rigorous.

And it, it will not be possible to make do full justice to this theory in this short span of time that we have. So, my goal here is to not go through all the rigorous mathematics. But, then only give you a high level idea of the important points in constrained optimization, with a specific focus on topics, which will be relevant to solving constrained optimization problems in the machine learning context.

For now we will look at it as a general problem, and towards the end of this lecture, I will point out how some machine learning problems actually use the theory that we will discuss in a big way. So, that would be the goal of this the next some maybe few minutes. So, how do we solve

this problem? It is not immediately obvious what the solution is going to be or anything of that sort.

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But, let us start by looking at this function, which I am going to call as the Lagrangian function. You may remember that we have discussed Lagrangians, when we looked at constrained optimization earlier. This would be similar to that; just that we are going to make it slightly more formal and rigorous. So, not in the full trigger, but then we will make it slightly more rigorous than what we looked at earlier; that was more like a geometric intuition.

Here is more like a proper formalism that we will look at. So, what is a Lagrangian function? A Lagrangian function can be defined as follows. At least for this function let us define it as a function L, which takes as inputs; not just one input but two inputs. Let us call this x and some lambda. So, notice that this x is a can be a vector, especially if your function f is from Rn to R; then this x is actually a vector.

This lambda, in this case is a scalar; and what is this lambda indicating? So, the lambda is somehow associated with the constrained that you have; and we will see in a minute. So, if we have more constrained, later on we will see you will have you will need more such scalars. But right now we have only one constraint, so one scalar would do.

So, what is this function? This function is going to be defined as follows; so, this is f of x + lambda times h of x. So, this is a new function that we are defining, given the original constrained optimization problem, which we want to understand how to solve. Now, what is the use of this function? So why do we care about this function. So, this is a very very interesting function because of the following reason.

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So, let us say we fix some x, some x in this could be in Rn in genera; now, and consider the following problem. Maximize over all lambda ≥ 0, L of x comma lambda. It seems like pulled this function out of the hat; this problem out of the hat. But we will see why this is very relevant here; for the moment let us just look at this problem. So now, notice that the moment I fixed an x, so this is just a function of lambda.

So, you give me an x, I will fix that x; and I will ask what is the value of L of x comma lambda; as I vary lambda and lambda can take any value which is ≥ 0. What is L? So, let me explicitly write this L; so this is just f of x + lambda h of x. Now, think for a minute, what what values can this particular function, take; let me ask the following question.

Let us say the x that they give you or the x that you gave me, satisfies a h of x is ≤ 0. So, if h of x is ≤ 0; now, and if I am trying to maximize f of x + lambda h of x. Now, how big can I make this value. Remember, our lambda is constrained to be non negative value; so it cannot take negative

values, it has to be positive or 0. h of x we are saying for the particular x that you give me; h of x is ≤ 0.

So, lambda is a non negative quantity, sorry about that. Lambda is a non negative quantity, h of x is ≤ 0; the product, therefore, has to be ≤ 0, it cannot be positive. So, which means that I have, you give me an x and f of x is fixed; so this is a problem with respect to lambda. So, this is you can treat f of x as a constant; now, the lambda appears only here. Now, the question is, how big can I make this as I change lambda?

Now, it is very clear that, adding any lambda or using any lambda which is strictly positive will only hurt me. So, because if h of x is in fact strictly less than 0; so making lambda positive will only make this product negative. Which means that the whole objective value will decrease as opposed to having lambda equals 0. If I put lambda equals to 0; then if h of x is less than 0, then what would happen?

What would happen is that the second term is going to be 0. We know that the second term. We know that the second term is lambda times h of x, which is non negative. So, I am sorry, which is, which cannot be positive; so, which means that the best possible value if you are trying to maximize it is 0. If h of x is already 0, then it does not matter what lambda; I put this whole thing is going to be, the second term is going to be 0.

But if h of x is less than 0, then because I have the power to choose any lambda ≥ 0; I will choose lambda equal to 0, such that this whole thing becomes f of x + 0. Which means what? So which means the conclusion is that, if h of x is less than 0; this maximization, the value at the maximum is simply going to be f of x. Because the second term can always be made to made to be 0; and it cannot be made positive at all.

Now, what would happen if h of x is actually greater than 0? So, pause for a minute and think about this. What would happen if h of x is greater than 0; maybe some of you are already seeing it. So, if h of x is positive, and I am allowed to pick any lambda which is ≥ 0; what would be the natural thing to do? I would pick lambda as big as possible.

The bigger I pick lambda, the product is going to become even more bigger; because h of x is anyway positive. It is going to become bigger as I increase lambda, and so I can increase lambda

infinitely, to make this value go as big as infinity. So, which means that when h of x is greater than 0, this is going to be infinity. Now, what is this is and very very interesting property of this function; so and why is that.

And that is because what we are saying is that here is a function, where if I maximize with respect to lambda ≥ 0. If my condition is satisfied, the original condition of the problem that they cared about, which is h of x is ≤ 0. If that condition is satisfied, then I get the functions exact value. If the condition is not satisfied, then I get infinity as the value. Which means what?

Which means that you can think of this as like, some set constrained set, where h of x is ≤ 0. So this is in this case, at f is from R2 to R, and and what, and h is also R2 to R; which means that your domain is just that plane. And what we are saying is that this is the region, where h of x is ≤ 0.

And now if I look at the functions value at this region; so in this region, for this new function that I have constructed, it is going to be exactly f of x. Whereas, the moment I go outside this region, if I consider an x outside this region; the value is going to be infinity. So, anywhere inside it is going to be f of x; anywhere outside it is going to be infinity.

Which means, this suggests that if I want to solve the original problem, which was minimize over x, f of x; such that h of x is ≤ 0. This is the original problem that they wanted to solve. Now, equivalently, I can also solve this following problem. Which is, I want to maximize lambda ≥ 0, L of x comma lambda; but then here I have fixed an x. Depending on whether x is inside the constrained or outside the constrained set; I am going to either get f of x or infinity.

So, the moment it goes outside the constrained set, the function becomes infinity; which means that I can minimize this function over all possible x. So, where is that minimum going to be? If x is outside the constrained set, the value is going to be infinity; so that cannot really be the minimum. If x is insane inside the constrained set, I get exactly f of x is the value; which means that, if I look at the minimum over all possible x, it will necessarily be the solution to the original problem that we started off with.

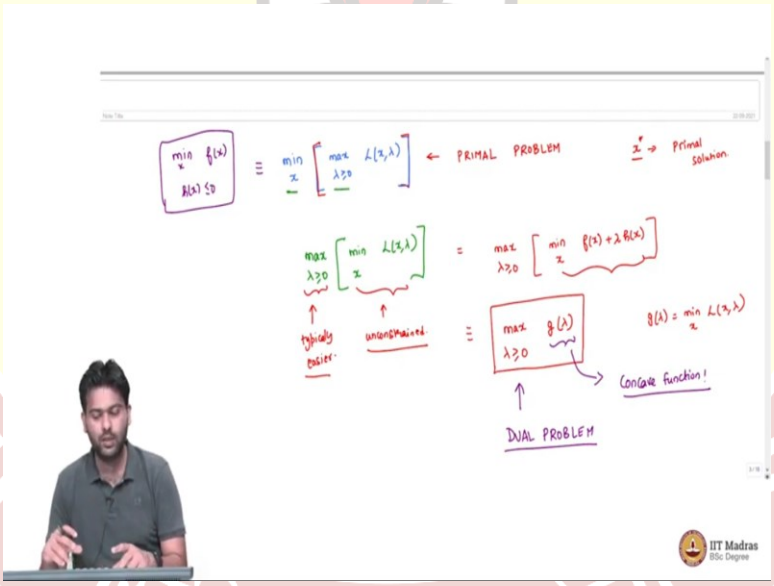
So this, these two are exactly equivalent problems; and that is why this Lagrangian is a very insightful function to look at. All we have said so far is that, I wanted to solve the original

problem; which had just f of x, such that h of x is ≤ 0. Now, it looks like I converted a minimization problem into a min-max problem.

So, I have made it minimization over x, maximization over lambda ≥ 0, L of x comma lambda. And I am saying that the solution to this problem, wherever that x* that I get. If I look at the L of x* comma lambda whatever; so, if I solve this problem, essentially I will get the same value as this problem; that is what I am saying right now. But that is still not telling us how to solve this problem.

So, this problem can be hard to solve; we have not made it any easier per se. So, we started with the problem, we changed it to a different formulation; and it does not look like it is any easier. Now, yes, this is probably not any easier. But why is this still a useful, very very useful idea is because of the following reason. So, now, one can ask.

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So, let me call this problem and let me read like this, because it is necessary. So, remember we are minimizing over x, maximizing over lambda ≥ 0, L of x comma lambda. Let me call this the primal problem; and let us call x* the primal solution. When I say x*, I am still thinking of this as a minimization problem over x.

So, when I fix an x, I get a value, and I am trying to minimize this over x basically. So, and call this x* as the primal solution. What we know about the primal solution, f of x* is the value at of

this function, evaluated at x* is exactly f of x*. So, where f of x* is the optimal value for the original problem that we wanted to solve; so this, let this be the primal solution.

So, now what we are going to do is look at a slightly modified version of this problem. So, what we are going to do, in particular, is something very interesting. The moment you see a min and max, just out of curiosity, we asked what would happen if we swap the min and max. In particular, I consider the following problem. Instead of min over x, max over lambda greater than 0; I now look at max over lambda greater than 0, min over x L of x comma lambda.

Now, I mean so, so it is not immediately obvious why this is going to be helpful, nor. So, it is not true that these two values are going to be the same. So, if you change min and max its, there is no guarantee that the value that you will get out of this optimization problem, is going to be the same as the other problem. If it is the same, then, I can solve this problem also. We do not know what at this point we do not know why that would be useful also.

But for the moment, we just for the sake of curiosity consider this new problem. So, what does this say? This say I am going to maximize over lambda ≥ 0, minimize over x L of x comma lambda. So, the interesting part of this problem is that, this part is an unconstrained optimization problem; this is an unconstrained optimization problem.

So, initially we had a constrained optimization problem, which is f of x minimize f of x, such that h of x is less than or equal 0. Now, this part inside the bracket is an unconstrained optimization problem. So, if I have to write this up explicitly, so this is max over lambda greater than 0, minimize over x, f of x + lambda h of x. And we know how to solve unconstrained optimization problem at least; so in principle, we that is we know that is easier to solve.

Now, of course that is not the only part; the other part is the maximization. This maximization is just over lambda ≥ 0; so, this is typically easier. So, I am not saying it is always going to be easier; but this constrained is slightly easier constrained, because it is just saying that all values of lambda are going to be ≥ 0.

So, as opposed to saying we want h of x is ≤ 0, which was the original constrained that we care about. We somehow converted that into max-min problem, where one part is unconstrained, the

other part has an unconstrained, which is typically easier to handle; now, something more is also true; So, now if I think of this as max over lambda ≥ 0, g of lambda.

If I think of this as a function, where g of lambda min over x, L of x comma lambda, this part this; whatever is inside the bracket, so let us call that g of lambda, a separate function. So, this is maximize over lambda $\geq$ g of lambda; now it, here is interesting part. So, this g of lambda happens to be a concave function; now it is a very interesting exercise to show that g of lambda is a concave function.

So, g what is g of lambda? g of lambda is minimization over x, L of x comma lambda or minimization over x, f of x + lambda h of x. So, given a lambda, the function g of lambda is going to map it to minimum of several linear functions of lambda. Infinitely linear functions of lambda, where these infinite functions comes from each x.

And we can show again from whatever properties that we looked at in the previous class for definition of convex functions, equivalently concave functions; we can show that g of lambda is concave function. It would be great exercise to show that this is a concave function, from whatever you learned about the definitions of convexity. Which means that we now have a problem which is a concave maximization problem, where the constrained is lambda ≥ 0; which is which is easier problem to solve than the original problem.

But, remember we swap the max and min. So, we do not know how that is going to affect the solution of the original problem that we cared about. The original problem just to recollect the original problem was min over f of x, h of x is ≤ 0. We argued that this guy is equivalent to the primal problem.

Now, what we have done is we have swapped the min and max in the primal problem; and we have gotten a different problem, which let me call that as dual problem. So, which is a maximization over lambda g of lambda problem. But, we do not know how solving a dual problem is going to help solve the primal problem; or, or are there any relationships at all.

So, it maybe true that the dual problem is easier to solve, it is a concave problem; it is a concave maximization problem, which are not, not too hard to constraints. The constraints are lambda $\geq$ 0; maybe it is easier to solve. But, just because it is easier to solve; it does not mean it is useful.

So, because our original goal was to solve the primal problem, we have looked at a related problem which is the dual problem, is seems easier to solve.

Now, the question is how are the primal and the dual related to each other? Because understanding that will give a lot of insights about how to solve the primal problem, which is equivalent to solving the original constrained problem that we carried out, care to solve initially. So, what we are going to see next is how to relate the primal and the dual problems.

